

Public Economics

Lecture 1 · Why Study Public Finance?

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Today's Questions

- What is **public finance**, and what are the key questions the field addresses?
- What are the key facts about the size and growth of government, and the distribution of taxes and spending — in the US, across the OECD, and here in **Austria**?
- How do the questions of public finance show up in a concrete policy debate — the Covid-19 pandemic?

Based on

Gruber, J. *Public Finance and Public Policy*, Ch. 1 — “Why Study Public Finance?”

A Motivating Case: Covid-19

1.1 The Four Questions of Public Finance

1.2 Facts on Government: US, OECD, and Austria

1.3 Back to Covid-19: The Four Questions in Practice

1.4 Conclusion

A Motivating Case: Covid-19

2020: A Natural Experiment in the Role of Government

United States

- GDP fell 31.2% (Q2 2020); unemployment rose to 14.8%
- \$2.2 trillion CARES Act (March 2020): direct payments, expanded UI, business loans
- States diverged sharply: strict lockdowns (CA, NY) vs. minimal restrictions (FL)
- Republicans vs. Democrats disagreed on *almost every* dimension of the response

Austria

- First EU country to enter a full nationwide lockdown (March 16, 2020)
- *Kurzarbeit* (short-time work) massively expanded to prevent layoffs
- COVID-19 relief packages: Fixkostenzuschuss, Ausfallsbonus, Corona-Hilfsfonds
- Feb 2022: Austria became the *first EU country* to legislate a general Covid-19 vaccination mandate

Every one of these decisions is a public finance decision.

1.1 The Four Questions of Public Finance

What Is Public Finance?

Public finance

The study of the role of **government** in the economy.

The four questions of public finance:

1. **When** should the government intervene in the economy?
2. **How** might the government intervene?
3. What is the **effect** of those interventions on economic outcomes?
4. **Why** do governments choose to intervene in the way that they do?

Q1: When Should the Government Intervene?

Baseline: in a competitive market, the equilibrium (supply = demand) maximizes the *social efficiency* of trade.

- Every trade valued by both sides happens; no valuable trade is missed.
- So the free market is the benchmark — government intervention needs a *justification*.

Two justifications for intervention

1. **Market failure** — the market outcome is not efficient
2. **Redistribution** — society wants a different *distribution* of the (efficient) outcome

Market Failure: The Health Insurance Example

Market failure

A problem that causes the market economy to deliver an outcome that does not maximize efficiency.

- Health insurance *looks* like a textbook competitive market: many buyers, many sellers.
- But: if I skip a flu shot, I raise *my* risk of getting the flu — and the risk that I infect *you*.
- I only weigh *my own* costs and benefits when deciding whether to insure/vaccinate; I ignore the cost I impose on others.

⇒ a **negative externality**: private decisions $\neq$ socially efficient decisions.

Application: Vaccines, Externalities, and Herd Immunity

- US measles: vaccine (1963) cut cases from 3–4 million/year to under 3,000/year by the 1980s.
- 1989–91 resurgence (50,000+ cases) and a 2014–15 Disneyland outbreak were traced to *unvaccinated* clusters.
- Response: California (2014) ended personal-belief exemptions for school enrollment — vaccination rates rose sharply.
- Worldwide: 200,000 measles deaths in 2019, a 50% rise in three years.

The logic

Because an unvaccinated person imposes costs on others, private choices under-provide vaccination relative to the social optimum — a textbook case for government intervention.

Austria's Case: The 2022 Covid-19 Vaccination Mandate

- Feb 2022: Austria's *Impfpflichtgesetz* made Covid-19 vaccination compulsory for adults — a first in the EU.
- Justified explicitly on **externality** grounds: protecting the health system and others, not just oneself.
- June 2022: the *Verfassungsgerichtshof* (Constitutional Court) upheld the law *in principle*, but applied strict scrutiny because a vaccine mandate is a serious intervention in bodily autonomy.
- The mandate was never enforced with fines and was repealed in 2023 as Omicron changed the risk calculus.

Why this matters for public finance

A textbook market-failure argument (Q1) still had to survive constitutional *and* political scrutiny (Q4) before becoming policy (Q2) — and it was reversed once the underlying facts changed (Q3).

The Second Motive: Redistribution

Redistribution

The shifting of resources from some groups in society to others.

- Think of the economy as a **pie**: efficiency determines its *size*.
- Society may also care about the *distribution* of slices — e.g., valuing €1 more to a poor household than to a rich one.

The equity–efficiency trade-off

Redistribution usually changes behavior (e.g., work incentives), which can *shrink the pie* even as it changes how it is sliced.

Why Redistribution Is on the Table in Austria

Income inequality

- Gini coefficient (equivalized disposable income): 30.7 in 2021, up from 29.8 in 2020
- At-risk-of-poverty rate: $\approx$ 12–14% of the population
- Material deprivation nearly doubled: 201,000 people (2.3%) in 2022 $\rightarrow$ 336,000 (3.7%) in 2023

Wealth inequality

- Median household net wealth: $\approx$ €83,000
- Top 1% hold roughly 40%+ of total net wealth
- Bottom 50% hold under 4% of total net wealth

Source: Statistik Austria, EU-SILC (2023); Oesterreichische Nationalbank, Household Finance and Consumption Survey (HFCS).

Q2: How Might the Government Intervene?

Four broad policy tools:

1. **Tax or subsidize** private sale/purchase (change the price)
2. **Restrict or mandate** private sale/purchase (change the quantity directly)
3. **Public provision** (the government produces the good itself)
4. **Public financing of private provision** (government pays; private firms produce)

The Four Tools, With US and Austrian Examples

Tool	US example	Austrian example
Tax/subsidize	ACA premium tax credits for low-income buyers	<i>Pendlerpauschale</i> (commuter tax deduction); CO ₂ pricing under the eco-social tax reform
Mandate/restrict	ACA individual mandate (since repealed)	Covid-19 vaccination mandate (2022, repealed 2023)
Public provision	Medicare (elderly), VA hospitals	ÖGK public health insurance; public hospitals (KAV/Gemeindespital network)
Public financing of private provision	Medicare Part D (private insurers, public funds)	Social-insurance reimbursement of private physicians and private hospitals

Q3: What Are the Effects of Interventions?

Direct effects

What would happen *if people did not change their behavior*.

Indirect effects

What happens *because* people change their behavior in response.

Worked example (Gruber): covering the uninsured in the 2010 US

- 49 million uninsured $\times$ \$2,500/year $\Rightarrow$ direct cost $\approx$ \$125 billion
- If half of the privately insured *drop* their coverage to join the free public plan: cost triples to $\approx$ \$340 billion
- If only 10% drop coverage: cost rises to $\approx$ \$165 billion

The indirect effect — how people respond — is an empirical question, and it is often the whole ballgame.

Application: Independent Scorekeepers

Answering Q3 requires credible, nonpartisan empirical analysis.

United States: the CBO

- Congressional Budget Office (est. 1975)
- “Scores” the cost of every major bill before it can pass
- Its 2017 estimate (30M+ more uninsured under ACA repeal) helped sink the repeal effort

Austria: the Fiskalrat & Budgetdienst

- *Fiskalrat* monitors compliance with EU and national fiscal rules
- Parliament's *Budgetdienst* independently costs and analyzes budget bills for MPs
- Both flagged the 2024 deficit overshoot well before the official numbers were finalized

Q4: Why Do Governments Do What They Do?

- Q2 was **normative** (how *should* government intervene?). Q4 is **positive** (why do governments actually behave as they do?).
- Governments aggregate the preferences of millions of citizens — imperfectly.
- Just as markets can fail, **governments can fail** too: policy can reflect lobbying, electoral incentives, or institutional constraints rather than pure efficiency or equity goals.

Political economy

The theory of how the political process produces decisions that affect individuals and the economy.

Same virus, same facts, very different policies in the US, Canada, Germany, the UK — and across Austria's own coalition governments.

1.2 Facts on Government: US, OECD, and Austria

Why Look at the Facts?

- Public finance is compelling because government is *everywhere* in economic life.
- We now look at five facts: size, decentralization, deficits/debt, spending composition, and revenue composition.
- For each, we compare the textbook US numbers to **Austria and the OECD** — the environment you will actually work and pay taxes in.

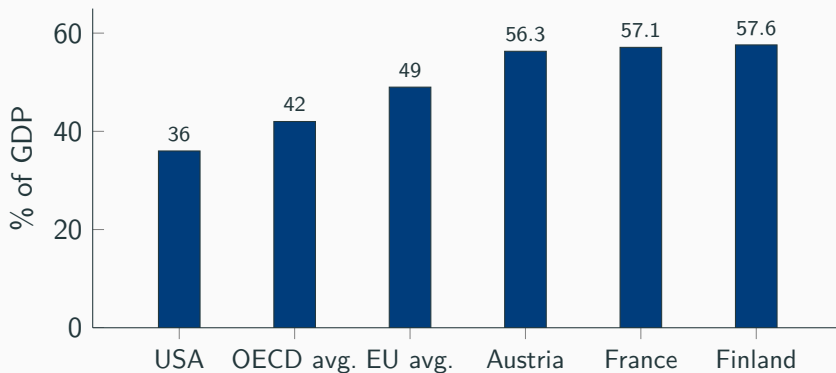
Fact 1: The Size of Government Has Grown — Everywhere

- US federal spending: 3.4% of GDP (1930) → $\approx 50\%$ of GDP (1944, WWII peak) → $\approx 20\%$ of GDP (recent decades)
- Growth has been even faster in many other OECD countries since 1960
- Greece: government share roughly tripled since 1960; Sweden peaked near two-thirds of GDP in the early 1990s

Discussion

What explains the twentieth-century growth of government spending across almost every developed country?

Austria Today: One of the Largest Governments in the EU



Austria had the **3rd-highest** total government spending ratio (“Staatsquote”) in the EU in 2024.

Source: Eurostat; Statistik Austria (2024); OECD Government at a Glance.

Fact 2: Decentralization — Who Spends, Who Taxes?

United States: the federal government provides the majority of spending, but state/local spending is still $>40\%$ of total government spending ($>17\%$ of GDP).

Austria's federalism paradox

- **Bund** (federal): collects $\approx 72\%$ of tax revenue, but does only $\approx 48\%$ of spending
- **Länder:** collect $\approx 12\%$ of tax revenue, but do $\approx 31\%$ of spending
- **Gemeinden** (municipalities): collect $\approx 12\%$ of tax revenue, but do $\approx 25\%$ of spending

Revenue-raising power and spending responsibility are separated — a classic “who pays vs. who decides” problem in fiscal federalism.

Source: Agenda Austria, fiscal federalism analysis; Austrian *Finanzausgleich*.

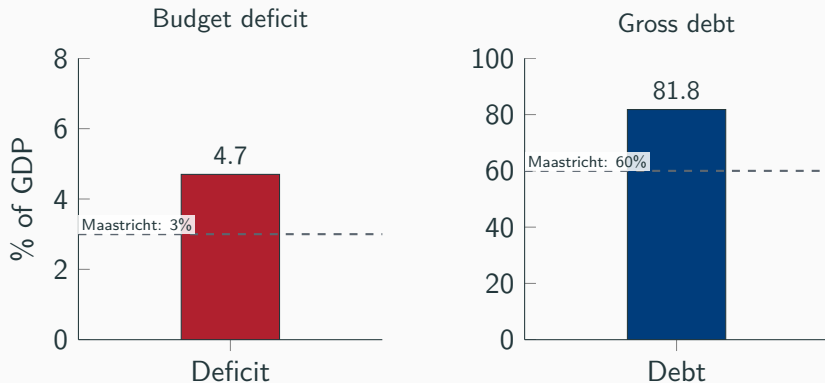
Fact 3: Deficits and Debt

Definitions

Deficit = spending — revenue in a given year. **Debt** = accumulated past deficits.

- US: near balance for decades (excl. WWII), large deficits since the 1970s–80s; debt now above 100% of GDP.
- State/local governments in the US are typically required to balance their budgets — and mostly do.

Austria vs. the Maastricht Rules



Austria 2024: deficit **4.7%** of GDP, gross debt **81.8%** of GDP — both above their Maastricht ceilings (3% and 60%).

Source: Statistik Austria, Maastricht notification (2025); BMF, Öffentliche Schulden 2024.

Discussion: Deficits and Debt

- What are the costs of running larger deficits and carrying more debt?
- Why might it make sense to let deficits rise in *some* years (recessions, pandemics) even under fixed fiscal rules?
- Austria's Länder and Gemeinden are far closer to balanced budgets than the Bund — why might that be, given Fact 2?

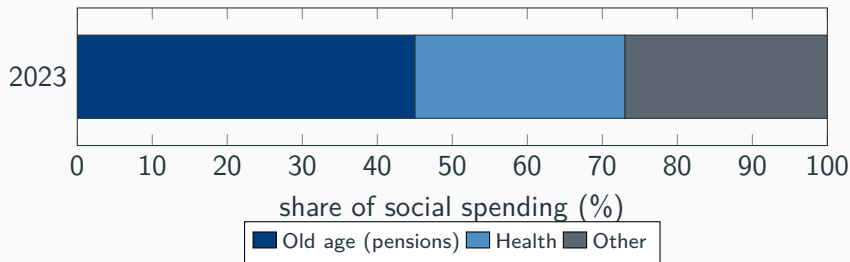
Fact 4: What Does Government Spend On?

US federal budget, then and now: defense fell from $\approx$ half of spending (1960) to under one-sixth today; Social Security ($\approx 19\%$) and health programs ($\approx 30\%$) grew to take its place.

Public goods vs. social insurance

Defense is a classic **public good** (non-excludable, non-rival benefit). Pensions and health coverage are **social insurance**: government provision of insurance where private insurance markets under-perform.

Austria's Social Spending



“Other” = family benefits, unemployment insurance, disability, and housing support.
Total social spending: €146.2 billion (2023). Austria's **Sozialquote**: 30.9% of GDP — among the highest social-spending ratios in the world.

Source: Statistik Austria, ESSPROS social protection statistics (2023).

Fact 5: Where Does the Revenue Come From?

United States: individual income tax ($\approx$ half of federal revenue) has stayed roughly constant; corporate tax fell from $\approx 25\%$ to $\approx 6\%$ of revenue; payroll taxes rose from one-sixth to over one-third.

Discussion

What are the implications of shifting the tax burden from businesses and consumption toward workers' earnings?

Austria's Tax Mix: Heavy on Labor

- VAT (*Umsatzsteuer*): $\approx$ €35.4 billion (2022) — the single largest tax
- Wage tax (*Lohnsteuer*): $\approx$ €25 billion — about one-sixth of total tax revenue
- Corporate tax (*Körperschaftsteuer*): $\approx$ €7.8 billion — only 2.2% of GDP
- VAT + wage tax alone $\approx$ two-thirds of all tax revenue

A distinctive feature

Austria taxes **labor** more heavily, relative to the size of its economy, than any other EU country — while capital and consumption taxes play a comparatively smaller role.

Source: Agenda Austria; Bundesministerium für Finanzen (BMF), tax revenue statistics (2022–24).

Fact 6: The Regulatory State

Beyond taxing and spending, governments **regulate**.

Domain	United States	Austria
Food & drug safety	FDA	AGES (Agentur für Gesundheit)
Workplace safety	OSHA	Arbeitsinspektorat, AUVA
Environment	EPA	Umweltbundesamt
Communications	FCC	RTR (Rundfunk und Telekom Regulierungs-GmbH)

Regulatory agencies can be tiny in budget (the FDA is $<0.025\%$ of the US budget) yet govern a huge share of everyday economic life.

1.3 Back to Covid-19: The Four Questions in Practice

The Four Questions, Applied

1. **When intervene?** Mask mandates, business closures, and vaccine mandates were all justified by *externalities* — exactly the logic of the flu-shot and measles examples.
2. **How intervene?** Compare Kurzarbeit (public financing of private employment) to US direct payments and PPP loans (public financing of private provision).
3. **What effect?** Hundreds of NBER working papers (23.8% of all 2020 output) tried to measure whether lockdowns, PPP, and UI expansions actually changed behavior.
4. **Why this policy?** Democrats and Republicans proposed very different follow-up packages *using the same evidence* — and Austria's own coalition partners split over mandate enforcement.

1.4 Conclusion

Highlights

- Government intervention is justified by **market failure** (grow the pie) or **redistribution** (reslice the pie) — not simply by good intentions.
- There are many tools to intervene: taxes/subsidies, mandates/restrictions, public provision, and public financing of private provision.
- Empirical public finance — direct vs. indirect effects — is essential to judging *how well* any policy actually works.
- Governments can fail too: political economy explains why actual policy often departs from the efficiency/equity ideal.
- Government in Austria is large by international standards (56.3% of GDP), fiscally federal in a distinctive way, and its tax base leans heavily on labor income.

Discussion Questions for Next Time

1. Is Austria's heavy reliance on labor taxation (vs. capital/consumption) efficient, equitable, both, or neither?
2. Austria's Bund raises 72% of tax revenue but does only 48% of spending. What are the risks of this mismatch between taxing and spending authority?
3. Consider the case for a wealth tax in a country where the top 1% hold $\approx 40\%$ of net wealth. Frame your answer using the equity–efficiency trade-off.
4. Was the 2022 Covid-19 vaccination mandate a case of correcting a market failure, or a case of government overreach? Use the four questions of public finance to structure your answer.

Textbook: Gruber, J. *Public Finance and Public Policy*, Ch. 1, Worth Publishers.

Austrian and international data:

- Statistik Austria: Staatsausgaben nach Aufgabenbereichen; Sozialquote/ESSPROS; EU-SILC income and living conditions
- Bundesministerium für Finanzen (BMF): Öffentliche Schulden 2024; tax revenue statistics
- Parlament Österreich, Budgetdienst: Gebarungserfolg und gesamtsstaatliche Entwicklung 2024
- Oesterreichische Nationalbank (OeNB): Household Finance and Consumption Survey (HFCS)
- Agenda Austria: Staatsquote, fiscal federalism, and tax-structure analyses
- Eurostat; OECD *Government at a Glance*
- Verfassungsgerichtshof: *Impfpflichtgesetz* decisions G 29/2022, G 37/2022

Thank you!

Questions and discussion